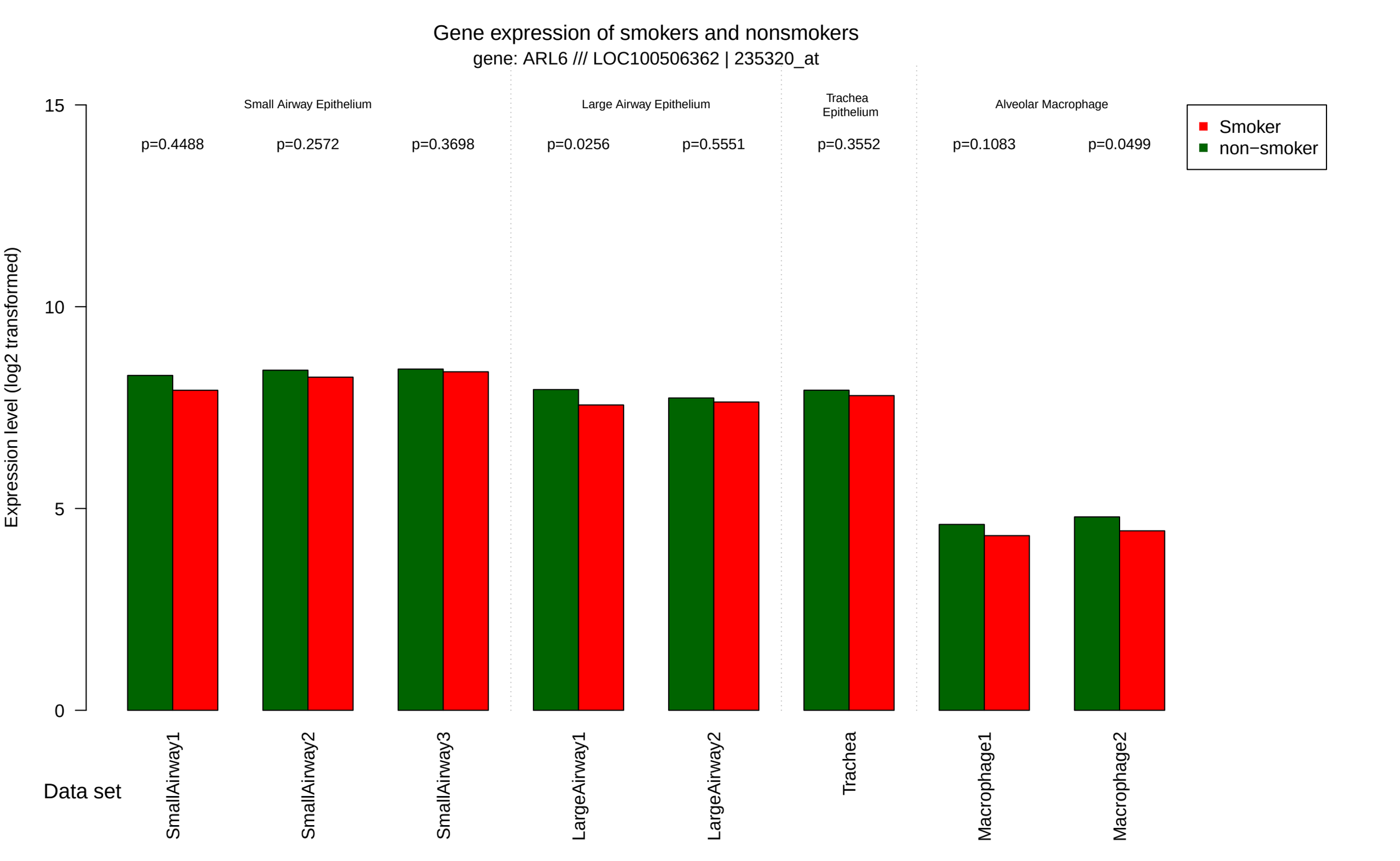
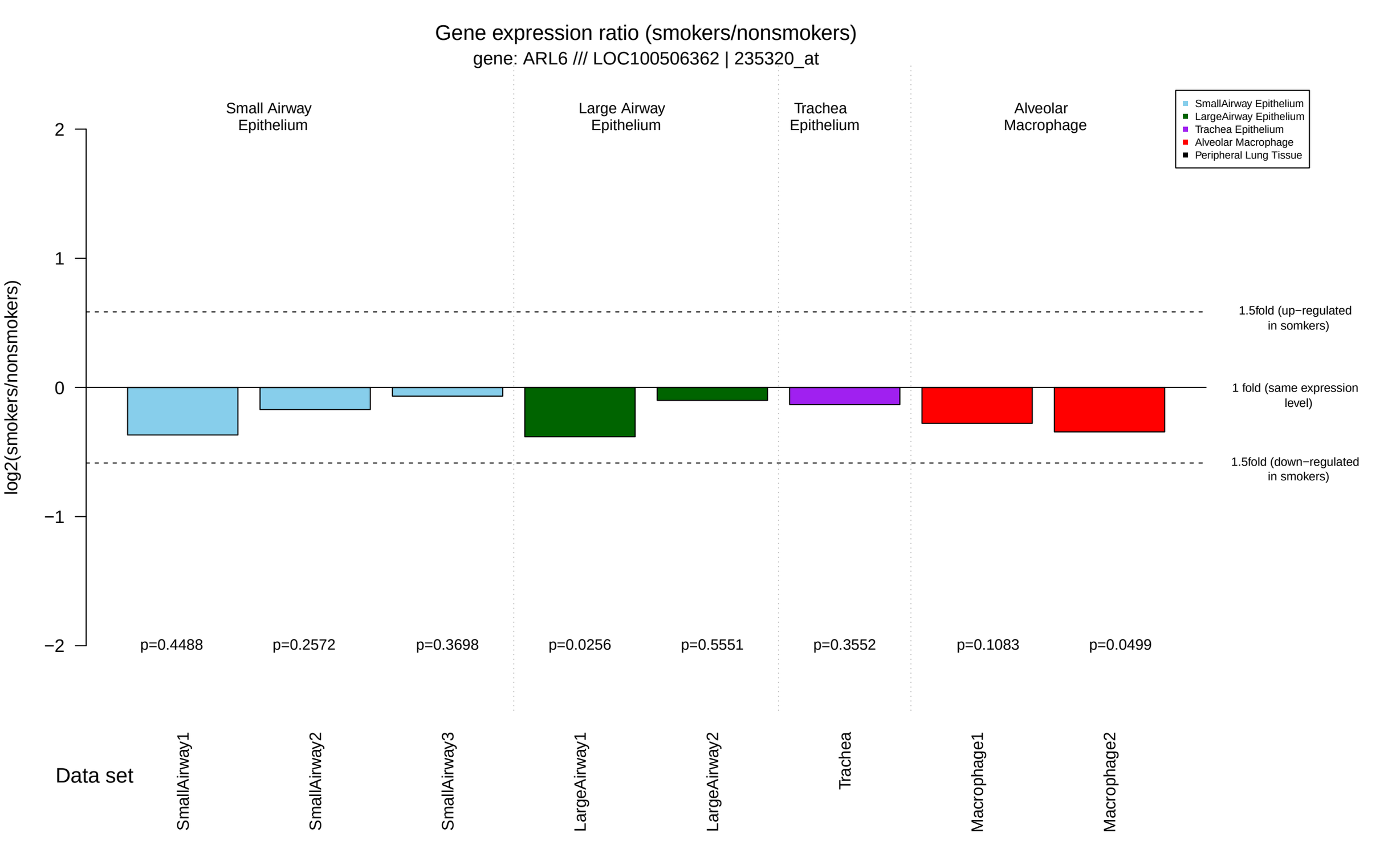
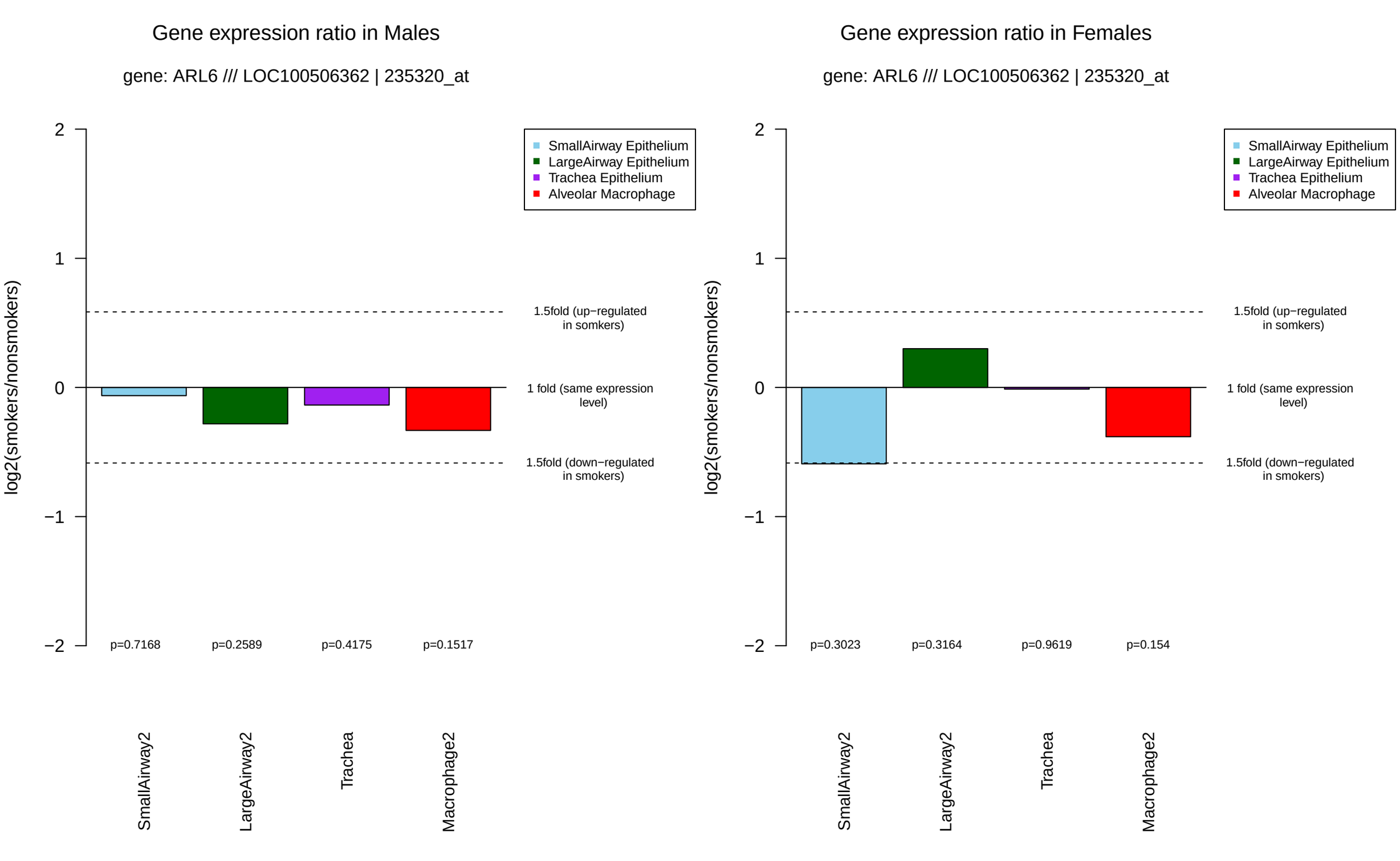
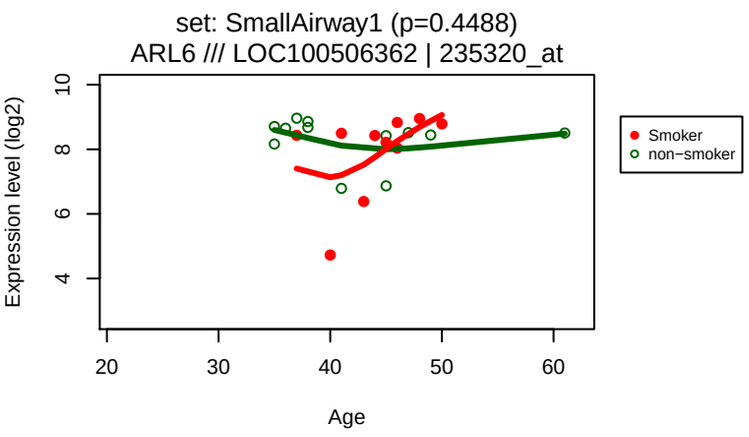
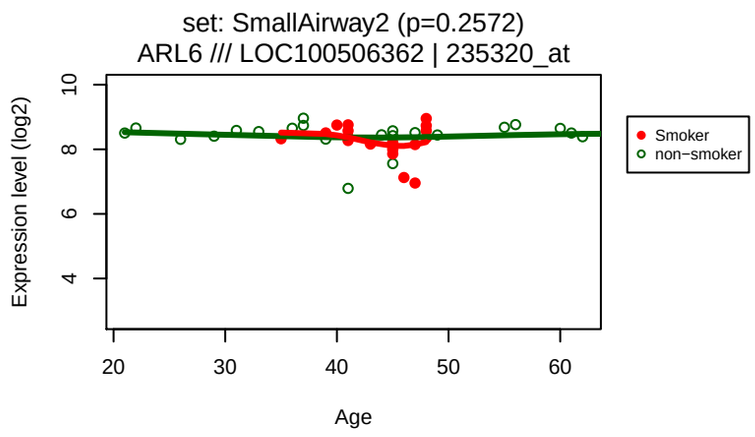
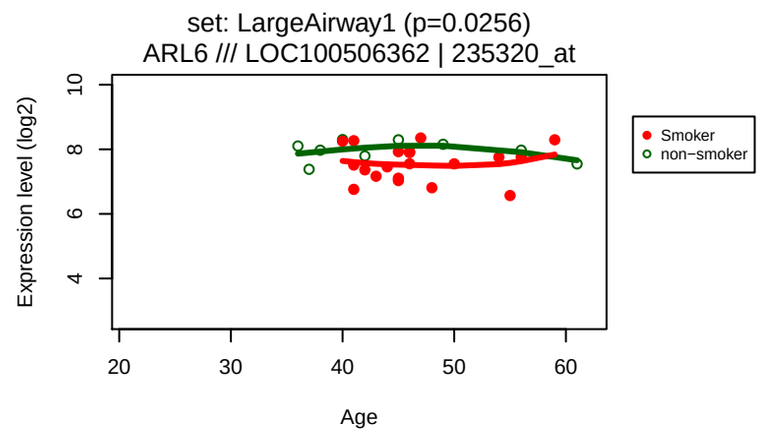

Gene expression of smokers and nonsmokers
gene: ARL6 /// LOC100506362 | 235320_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4488
p=0.2572
p=0.3698
p=0.0256
p=0.5551
p=0.3552
p=0.1083
p=0.0499
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

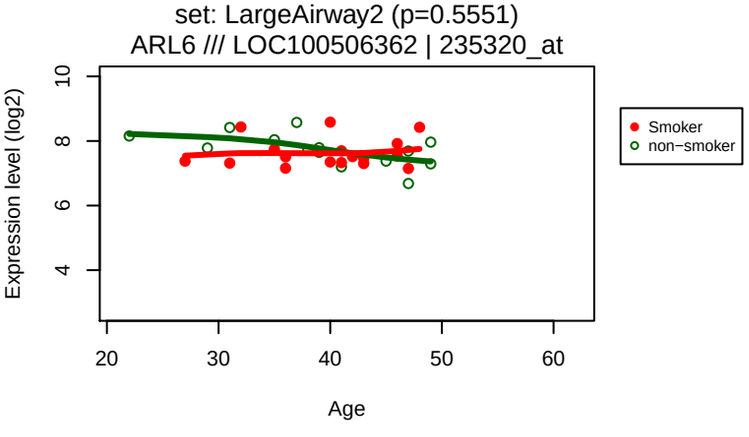

Gene expression ratio (smokers/nonsmokers)
gene: ARL6 /// LOC100506362 | 235320_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4488
p=0.2572
p=0.3698
p=0.0256
p=0.5551
p=0.3552
p=0.1083
p=0.0499
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

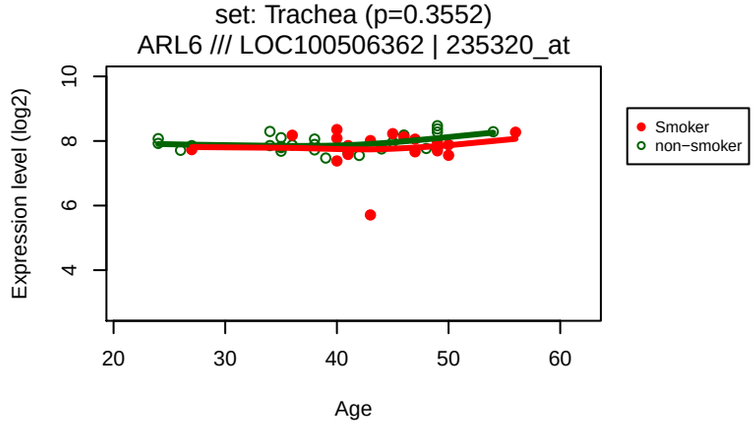

Gene expression ratio in Males
gene: ARL6 /// LOC100506362 | 235320_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7168
p=0.2589
p=0.4175
p=0.1517
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL6 /// LOC100506362 | 235320_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3023
p=0.3164
p=0.9619
p=0.154
SmallAirway2
LargeAirway2
Trachea
Macrophage2

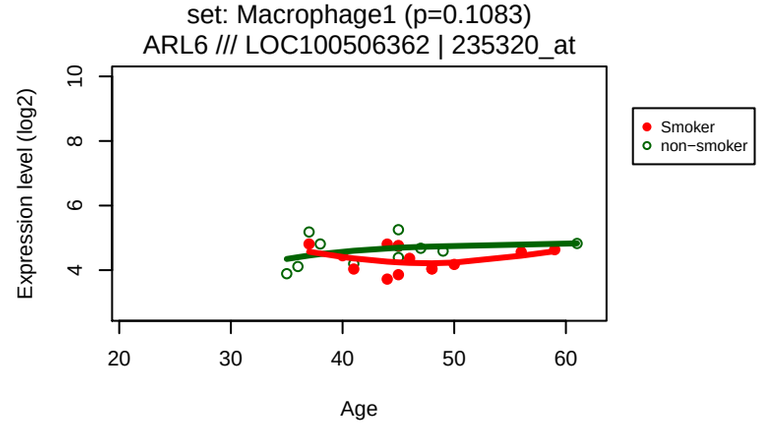

set: SmallAirway1 (p=0.4488)
ARL6 /// LOC100506362 | 235320_at
Expression level (log2)
Age
Smoker
non−smoker

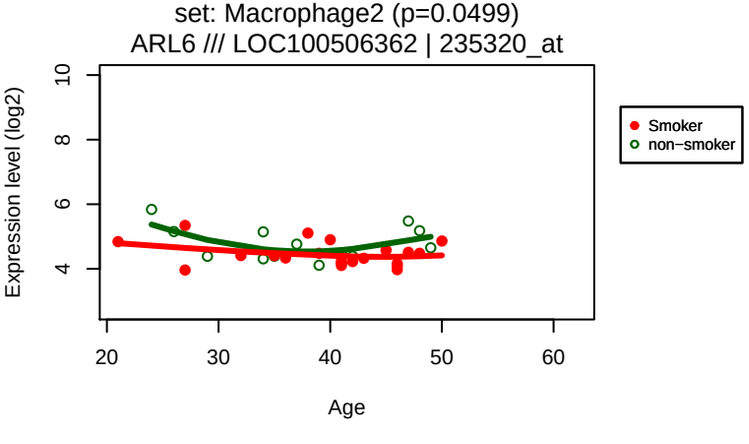

set: SmallAirway2 (p=0.2572)
ARL6 /// LOC100506362 | 235320_at
Expression level (log2)
Age
Smoker
non−smoker

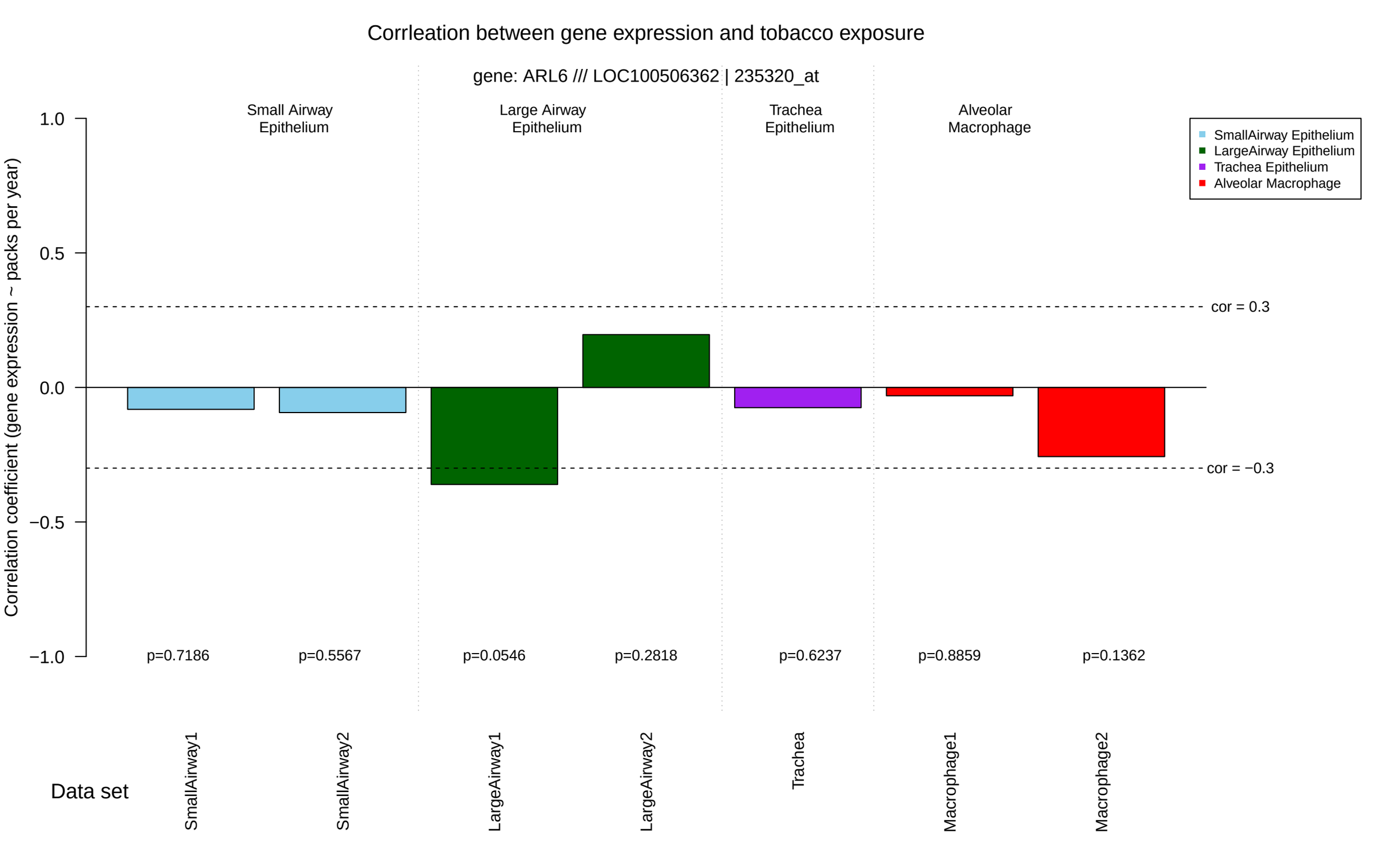

set: LargeAirway1 (p=0.0256)
ARL6 /// LOC100506362 | 235320_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5551)
ARL6 /// LOC100506362 | 235320_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3552)
ARL6 /// LOC100506362 | 235320_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1083)
ARL6 /// LOC100506362 | 235320_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0499)
ARL6 /// LOC100506362 | 235320_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL6 /// LOC100506362 | 235320_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7186
p=0.5567
p=0.0546
p=0.2818
p=0.6237
p=0.8859
p=0.1362
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2